

Annex ZA (normative)

Normative references to international publications with their corresponding European publications

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

NOTE When an international publication has been modified by common modifications, indicated by (mod), the relevant EN/HD applies.

<u>Publication</u>	<u>Year</u>	<u>Title</u>	<u>EN/HD</u>	<u>Year</u>
IEC 60076-2	-	Power transformers - Part 2: Temperature rise for liquid-immersed transformers	EN 60076-2	-
IEC 60076-3	2000	Power transformers - Part 3: Insulation levels, dielectric tests and external clearances in air	EN 60076-3	2001
IEC 60076-5	2006	Power transformers - Part 5: Ability to withstand short circuit	EN 60076-5	2006
IEC 60076-10	2001	Power transformers - Part 10: Determination of sound levels	EN 60076-10	2001
IEC 60076-11	2004	Power transformers - Part 11: Dry-type transformers	EN 60076-11	2004
IEC 60137	2008	Insulated bushings for alternating voltages above 1 000 V	EN 60137	2008
IEC 60214-1	2003	Tap-changers - Part 1: Performance requirements and test methods	EN 60214-1	2003
IEC 60296	2003	Fluids for electrotechnical applications - Unused mineral insulating oils for transformers and switchgear	EN 60296 + corr. September	2004 2004
IEC 60721-3-4	1995	Classification of environmental conditions - Part 3: Classification of groups of environmental parameters and their severities - Section 4: Stationary use at non-weatherprotected locations	EN 60721-3-4	1995
ISO 9001	2008	Quality management systems - Requirements	EN ISO 9001	2008

CONTENTS

FOREWORD.....	5
1 Scope.....	7
2 Normative references	8
3 Terms and definitions	8
3.1 General.....	8
3.2 Terminals and neutral point.....	9
3.3 Windings	10
3.4 Rating	11
3.5 Tappings	13
3.6 Losses and no-load current	15
3.7 Short-circuit impedance and voltage drop.....	16
3.8 Temperature rise.....	17
3.9 Insulation	17
3.10 Connections	17
3.11 Test classification.....	18
3.12 Meteorological data with respect to cooling	19
3.13 Other definitions.....	19
4 Service conditions	20
4.1 General.....	20
4.2 Normal service conditions	20
5 Rating and general requirements.....	22
5.1 Rated power.....	22
5.1.1 General	22
5.1.2 Preferred values of rated power.....	22
5.1.3 Minimum power under alternative cooling modes.....	22
5.1.4 Loading beyond rated power.....	23
5.2 Cooling mode	23
5.3 Load rejection on transformers directly connected to a generator	23
5.4 Rated voltage and rated frequency.....	23
5.4.1 Rated voltage	23
5.4.2 Rated frequency	23
5.4.3 Operation at higher than rated voltage and/or at other than rated frequency	24
5.5 Provision for unusual service conditions.....	24
5.6 Highest voltage for equipment U_m and dielectric tests levels.....	25
5.7 Additional information required for enquiry	25
5.7.1 Transformer classification.....	25
5.7.2 Winding connection and number of phases.....	25
5.7.3 Sound level	26
5.7.4 Transport.....	26
5.8 Components and materials	26
6 Requirements for transformers having a tapped winding.....	27
6.1 General – Notation of tapping range.....	27
6.2 Tapping voltage – tapping current. Standard categories of tapping voltage variation. Maximum voltage tapping.....	27
6.3 Tapping power. Full-power tappings – reduced-power tappings.....	30